



Individual Assessment Coversheet

To be attached to the front of the assessment.

Campus: Bedfordview

Faculty: Information Technology

Module Code: ITEJA0-11

Group: 1

Lecturer's Name: Thendo Siphuma

Student Full Name: Plamedi Kilangalanga

Student Number: EDUV10913744

Indicate	Yes	No
Plagiarism report attached		x

Declaration:

I declare that this assessment is my own original work except for source material explicitly acknowledged. I also declare that this assessment or any other of my original work related to it has not been previously, or is not being simultaneously, submitted for this or any other course. I am aware of the AI policy and acknowledge that I have not used any AI technology to generate or manipulate data, other than as permitted by the assessment instructions. I also declare that I am aware of the Institution's policy and regulations on honesty in academic work as set out in the Conditions of Enrolment, and of the disciplinary guidelines applicable to breaches of such policy and regulations.

Signature Plamedi Kilangalanga	Date 29 March 2026
--	------------------------------

Lecturer's Comments:

--

Marks Awarded:	%
-----------------------	---

Signature	Date
------------------	-------------

Eduvos (Pty) Ltd. (formerly Pearson Institute of Higher Education) is registered with the Department of Higher Education and Training as a private higher education institution under the Higher Education Act, 101, of 1997. Registration Certificate number: 2001/HE07/00

Section 1: Object-Oriented Design (Core Classes)

1.1 The Student Class

```
package eduvosonboardinggame;

public class Student {
    private String name;
    private int energy;
    private int academicScore;

    public Student(String name, int energy, int academicScore){
        this.name = name;
        this.energy = 70;
        this.academicScore = 0;
    }
    public void updateEnergy(int value){
        this.energy += value;
    }
    public void updateScore(int value){
        this.academicScore += value;
    }
    public String getName(){
        return name;
    }
    public int getEnergy(){
        return energy;
    }
    public int getAcademicScore(){
        return academicScore;
    }
}
```

1.2 The Scenario Abstract Class

```
package eduvosonboardinggame;
```

```
public abstract class classScenario {  
    public abstract void playclassScenario(Student student);  
  
}
```

1.3 The Scenario Subclasses

RegistrationScenario:

```
package eduvosonboardinggame;

import java.util.Scanner;

public class registration extends classScenario{

    @Override
    public void playclassScenario(Student student) {
        Scanner scanner = new Scanner(System.in);
        int choice;

        System.out.println ("=====Welcome to Day 1: Registraton
day!=====");

        System.out.println ("1.Arrive early (+10 score, -10 energy)");
        System.out.println("2. Arrive late (+5 score, -5 energy)");

        while(true){
            try{
                System.out.println("Enter your choice (1 or 2): ");
                choice = Integer.parseInt(scanner.nextLine());

                if (choice ==1){
                    student.updateScore(10);
                    student.updateEnergy(-10);
                    return;
                } else if (choice ==2){
                    student.updateScore(5);
                    student.updateEnergy(-5);
                    return;
                } else {
                    System.out.println("Invalid selection! choose either 1 Or
2.");
                }
            } catch (Exception e){
```

```

        System.out.println("INVALID INPUT! PLEASE ENTER A NUMBER.");
    }
}
}

```

LectureScenario:

```

package eduvosonboardinggame;

import java.util.Scanner;

public class LectureScenario extends classScenario {

    @Override
    public void playclassScenario(Student student){
        Scanner input = new Scanner(System.in);
        int choice;

        System.out.println("=====Welcome to day 2: Lecture 1! =====
");

        System.out.println("1. Attend lecture (+15 score, -15 energy)");
        System.out.println("2. Arrive late (0 score, +5 energy)");
        System.out.println("3. Bunk lecture (0 score, +10 energy)");

        while (true){
            try{
                System.out.println("Enter your choice (1, 2, or 3):");
                choice = Integer.parseInt(input.nextLine());

                switch (choice) {
                    case 1:
                        student.updateScore(15);
                        student.updateEnergy(-15);

```

```

        return;
    case 2:
        student.updateEnergy(5);
        return;
    case 3:
        student.updateEnergy(10);
        return;
    default:
        System.out.println ("Invalid selection! choose 1, 2
or 3.");
        break;
    }
} catch (Exception e){
    System.out.println("INPUT NUMBERS ONLY! TRY AGAIN.");
    input.next();
}

}

}
}

```

AssignmentScenario:

```

package eduvosonboardinggame;

import java.util.Scanner;

public class Assignments extends classScenario {

    @Override
    public void playclassScenario(Student student) {
        Scanner input = new Scanner(System.in);
    }
}

```

```

        int choice;

        System.out.println("=====Welcome to Day 3: Assignmet
day!=====");

        System.out.println("1.Begin assignment (+20 score, -20
energy)");

        System.out.println("2.Postpone assignment(+5 score, -5
energy)");

        while(true){
            try{
                System.out.println("Enter your choice (1 or 2): ");
                choice = Integer.parseInt(input.nextLine());

                if (choice == 1){
                    student.updateScore(20);
                    student.updateEnergy(-20);
                    return;
                } else if (choice == 2){
                    student.updateScore(5);
                    student.updateEnergy(-5);
                    return;
                }
                else {
                    System.out.println("CHOOSE EITHER 1 OR 2.");
                }
            } catch (Exception e){
                System.out.println ("INVALID INPUT! ENTER A
NUMBER.");
            }
        }
    }
}

```

StudyGroupScenario:

```
package eduvosonboardinggame;

import java.util.Scanner;

public class StudyGroup extends classScenario{
    @Override
    public void playclassScenario(Student student){
        Scanner input = new Scanner(System.in);

        while(true){
            try{
                System.out.println("=====Welcome to Day 4: Study Group
day!=====");
                System.out.println("1.Join study group (+15 score, -10
energy)");
                System.out.println("2.Self study (+10 score, -5 energy)");
                System.out.println("Choose option 1 or 2: ");

                int choice = input.nextInt();

                switch (choice) {
                    case 1 -> {
                        student.updateScore(15);
                        student.updateEnergy(-10);
                        return;
                    }
                    case 2 -> {
                        student.updateScore(10);
                        student.updateEnergy(-5);
                        return;
                    }
                }
            }
        }
    }
}
```



```

        default -> System.out.println("Invalid option! Try
again.");
    }
    } catch (Exception e){
        System.out.println("INSERT NUMBERS ONLY.");
    }
}
}
}
}
}

```

TestScenario:

```

package eduvosonboardinggame;

import java.util.Scanner;

public class Test extends classScenario{

    @Override
    public void playclassScenario(Student student){
        Scanner scanner = new Scanner(System.in);
        int choice;

        System.out.println("=====Welcome to day 5: Test
Day!=====");

        System.out.println("1. Write test prepared: (+25 score, -20
energy)");

        System.out.println("2. Write test unprepared: (+10 score, -10
energy)");

        while(true){

```

```

try{
    System.out.println("Choose option 1 or 2: ");
    choice = Integer.parseInt(scanner.nextLine());

    switch (choice) {
        case 1:
            student.updateScore(25);
            student.updateEnergy(-20);
            return;

        case 2:
            student.updateScore(10);
            student.updateEnergy(-10);
            return;

        default: System.out.println("Invalid option! Try
again.");
    }

    } catch (Exception e) {
        System.out.println("INSERT NUMBERS ONLY.");
    }

    }
}

```

Section 2: Application Flow & Execution

2.1 The MainApp Class

```
package eduvosonboardinggame;
import java.util.Scanner;
public class EduvosOnboardingGame {

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String studentName;

        System.out.println("=====
        =====");
        System.out.println("Get ready to level up!"
                           + "Welcome to the Eduvos- First Week simulation!");

        System.out.println("=====
        =====");
        System.out.println("Let's set up your profile!");
        System.out.println();

        //Step 1:Player Setup
        System.out.println("Enter your student name: ");

        while(true){
            studentName = scanner.nextLine();

            //Validating the name
            if (studentName.matches("[a-zA-Z]+")){
                break;
            }else{
                System.out.println("Invalid name! Please use letters only:
");
```

```
    }  
}
```

```
    System.out.println("Welcome, "+ studentName + "! Let's make this an  
incredible first week!");
```

```
//Step 2: Initialise student attributes
```

```
    Student student = new Student(studentName,70,0);
```

```
    System.out.println("Your starting status: ");  
    System.out.println("Student: "+ student.getName());  
    System.out.println("Energy: " + student.getEnergy());  
    System.out.println("Academic Score: " + student.getAcademicScore());
```

```
//Step 3: Scenarios (Day 1-5)
```

```
    classScenario[] scenarios = {  
        new registration(),  
        new LectureScenario(),  
        new Assignments(),  
        new StudyGroup(),  
        new Test()  
    };
```

```
    for (int day = 1; day <= 5; day++){
```

```
        System.out.println("=====
```

```
        System.out.println("DAY" + day);
```

```
        System.out.println("=====
```

```

        System.out.println("Student: " + student.getName());
        System.out.println("Energy: " + student.getEnergy());
        System.out.println("Academic Score: " + student.getAcademicScore());

System.out.println("=====
=====");

        scenarios[day - 1].playclassScenario(student);

        if (student.getEnergy() <= 0){
            System.out.println("You have burned out! Energy depleted!");
            System.out.println("GAME OVER.");
            return;
        }
    }

//RESULTS

System.out.println("=====
=====");

        System.out.println("                                END OF SIMULATION
");

System.out.println("=====
=====");

        System.out.println("Final Score: " + student.getAcademicScore());

        int score = student.getAcademicScore();

        if (score <=40) {
            System.out.println(" POOR PERFORMANCE! Hit the books hard and
make it to every class");
        }

```

```
        else if (score <= 60){
            System.out.println(" AVERAGE PERFORMANCE! Great first attempt, keep
going!");
        }

        else {
            System.out.println(" EXCELLENT PERFORMANCE!");
            System.out.println("YOU REALLY CRUSHED IT THIS WEEK!");
            System.out.println("Hats off to you!");
        }

System.out.println("=====");
        System.out.println("                Thank you for playing!
");

System.out.println("=====");
    }
}
```

Section 3: Programming Constructs Demonstration

Instructions: To ensure you receive full marks for the technical guidelines and validation requirements, paste the specific, isolated code snippets from your project where you successfully implemented the following.

3.1 Regular Expression Validation

```
while(true){
    studentName = scanner.nextLine();

    //Validating the name
    if (studentName.matches("[a-zA-Z]+")){
        break;
    }else{
        System.out.println("Invalid name! Please use letters only: ");
    }
}
```

3.2 Try-Catch Exception Handling

```
try{
    System.out.println("Enter your choice (1 or 2): ");
    choice = Integer.parseInt(scanner.nextLine());
}
```

3.3 Looping Constructs

```
while (true){
    try{
        System.out.println("Enter your choice (1, 2, or 3):");
        choice = Integer.parseInt(input.nextLine());
    }
}
```

```
for (int day = 1; day <= 5; day++){
```

3.4 Burnout Condition Check

```
if (student.getEnergy() <= 0){
    System.out.println("You have burned out! Energy depleted!");
    System.out.println("GAME OVER.");
    return;
}
```

3.5 Final Performance Evaluation

```
if (score <=40) {
    System.out.println(" POOR PERFORMANCE! Hit the books hard and make it to every class");
}

else if (score <= 60){
    System.out.println(" AVERAGE PERFORMANCE! Great first attempt, keep going!");
}

else {
    System.out.println(" EXCELLENT PERFORMANCE!");
    System.out.println("YOU REALLY CRUSHED IT THIS WEEK!");
    System.out.println("Hats off to you!");
}

System.out.println("=====");
System.out.println("                Thank you for playing!                ");
System.out.println("=====");
```


OUTPUT

```
run:
=====
Get ready to level up!Welcome to the Eduvos- First Week simulation!
=====
Let's set up your profile!

Enter your student name:
Sharon
Welcome, Sharon! Let's make this an incredible first week!
Your starting status:
Student: Sharon
Energy: 70
Academic Score: 0
=====
                        DAY1
=====

Student: Sharon
Energy: 70
Academic Score: 0
=====
=====Welcome to Day 1: Registraton day!=====
1.Arrive early (+10 score, -10 energy)
2. Arrive late (+5 score, -5 energy)
Enter your choice (1 or 2):
1
=====
                        DAY2
=====

Student: Sharon
Energy: 60
Academic Score: 10
=====
=====Welcome to day 2: Lecture 1! =====
1. Attend lecture (+15 score, -15 energy)
```

```

2. Arrive late (0 score, +5 energy)
3. Bunk lecture (0 score, +10 energy)
Enter your choice (1, 2, or 3):
1
=====
                                DAY3
=====
Student: Sharon
Energy: 45
Academic Score: 25
=====
=====Welcome to Day 3: Assignmet day!=====
1.Begin assignment (+20 score, -20 energy)
2.Postpone assignment(+5 score, -5 energy)
Enter your choice (1 or 2):
1
=====
                                DAY4
=====
Student: Sharon
Energy: 25
Academic Score: 45
=====
=====Welcome to Day 4: Study Group day!=====
1.Join study group (+15 score, -10 energy)
2.Self study (+10 score, -5 energy)
Choose option 1 or 2:
1
=====
                                DAY5
=====
Student: Sharon
Energy: 15
Academic Score: 60
=====

=====
                                DAY5
=====
Student: Sharon
Energy: 15
Academic Score: 60
=====
=====Welcome to day 5: Test Day!=====
1. Write test prepared: (+25 score, -20 energy)
2. Write test unprepared: (+10 score, -10 energy)
Choose option 1 or 2:
1
You have burned out! Energy depleted!
GAME OVER.
BUILD SUCCESSFUL (total time: 11 seconds)

```

References

Bloch, J., 2018. *Effective Java*. 3rd edition ed. Boston: Addison-Wesley.

Deitel, P & Deitel.H, 2020. *Java:How to Program*. 11th edition ed. Harlow: Pearson Education.

Oracle, 2024. *Java Documentation*. [Online]
Available at: <https://docs.oracle.com/en/java/>
[Accessed 28 March 2026].

Schildt, 2019. *Java:The Complete Reference*. 11th edition ed. New York: McGraw-Hill Education.

W3Schools, 2026. *Java Tutorial*. [Online]
Available at: <https://www.w3schools.com/java/>
[Accessed 28 March 2026].